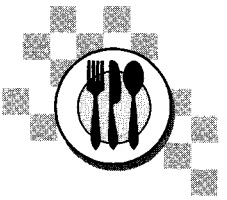
CHAPTER 2



Microorganisms

About This Chapter

In this chapter you will learn:

1. What microorganisms are and how they grow
2. How temperature, oxygen, water, pH, nutrients, and inhibitors affect the growth of microorganisms
3. How microbes cause food to spoil
4. About the causes and symptoms of the most common types of foodborne illness
5. How to destroy microbes
6. How to inhibit or control growth of microbes
7. About the tests that sanitarians use to measure the amount and type of microbes in food or on surfaces

Introduction

To understand food sanitation, you need to understand microorganisms that cause food spoilage and foodborne illness. Microorganisms are everywhere. Food spoilage happens when microorganisms spoil the color, flavor, or texture of food, and foodborne illness happens when people eat harmful microorganisms or their toxins along with the food. Good sanitation can prevent food spoilage and food-poisoning microorganisms from growing and causing damage.

Common Microorganisms

What Are Microorganisms?

A microorganism is a microscopic form of life found on anything that has not been sterilized. Microorganisms use nutrients, get rid of waste products, and reproduce.

About 100 years ago, Louis Pasteur and his coworkers realized that microorganisms cause disease and found that heat could destroy them.

Microorganisms and Food

Most foods spoil easily because they have the nutrients that microbes need to grow. Food is acceptable for a longer time, and foodborne illness is less likely, if the growth of microbes is controlled. Proper sanitation during food processing, preparation, and serving controls food spoilage and pathogenic (disease-causing) bacteria.

The most common microorganisms in food are bacteria and fungi. Fungi (molds or yeasts) are less common than bacteria. Molds are multicellular (organisms with several cells), and yeasts are usually unicellular (single-cell organisms). Bacteria usually crowd out fungi and are unicellular. Viruses are usually carried from person to person rather than via food, but they can be transferred among unhealthy employees. Parasites are not common in processed or cooked foods. While they can be foodborne, they do not multiply in food.

Molds

Molds have a variety of colors and look fuzzy or cottonlike. Molds can develop tiny spores that ride in the air and are spread by air currents. If these land on a new food and the conditions allow the spores to germinate, they grow. Molds can survive wider ranges of temperature and pH than bacteria and yeasts . Molds prefer pH 7.0 (neutral), but can tolerate pH 2.0 (acid) to 8.0 (alkaline). Molds prefer room temperature to refrigerator temperatures, but can even grow below freezing temperature. Most molds prefer a minimum water activity (A_w) of about .90, but some molds can grow at A_w as low as .60. (See later in this chapter for more about water activity.) At an A_w of .90 or higher, bacteria and yeasts usually grow better and crowd out molds. When the A_w is below .90, molds are more likely to grow than are bacteria and yeasts. Pastries, cheese, and nuts are low in water content and are more likely to spoil from mold than from bacteria and yeasts.

Yeasts

Yeasts have larger cells than bacteria and divide by producing buds. They multiply more slowly than bacteria: each yeast generation takes about 2 to 3 hours. It takes about 40 to 60 hours for food to be spoiled if it starts with one yeast organism per gram of food. Like molds, yeasts can be spread by air and also by hands, equipment, and surfaces. Yeast colonies usually look moist or slimy and are a creamy white color. Yeasts prefer an A_w of .90 to .94, but can grow below .90. A few can even grow at an A_w as low as .60. Yeasts like a somewhat acid pH, between 4.0 and 4.5. Yeasts are most likely to grow on vacuum-packaged foods with an acid pH. Growth of yeasts often makes foods smell fruity and feel slippery.

Bacteria

Bacteria are about 1 μm in diameter (0.00004 in). Their shape may be short rods (bacilli), round (cocci), or oval. Different types of bacteria group together in different ways. Some round bacteria form clusters like a bunch of grapes, e.g., staphylococci. Some rods or spheres link up to form chains, e.g., streptococci. Some round bacteria join together in pairs (diploid formation), e.g., pneumococci. Some form a group of four (tetrad formation), e.g. sarcinia. Bacteria make various pigments that may be yellow, red, pink, orange, blue, green, purple, brown, or black. Bacteria may discolor foods, especially those that have unstable color pigments, such as meat. Some bacteria discolor food by forming slime. Some bacteria produce spores that can survive heat, chemicals, and other harsh treatments. Some of these bacteria produce toxins that can cause foodborne illness.

Viruses

Viruses are infectious and are one-tenth to one-hundredth the size of bacteria. Most viruses can be seen only by using an electron microscope. Viruses can only grow inside the cells of another organism and live inside bacteria, fungi, plants, and animals. One virus that has caused several outbreaks of disease in restaurants in the past 10 years is hepatitis. Infectious hepatitis A can be carried by food that has been poorly handled. The symptoms are nausea, cramps, and, sometimes, jaundice, and can last from several weeks to several months. Raw shellfish from polluted waters are a common source of hepatitis. Foods that are handled a lot and those that are not heated after they are handled—such as sandwiches, salads, desserts, and ice—are most likely to transmit viruses. Hepatitis A is very contagious, so food handlers must wash their hands thoroughly after using the toilet, before handling food and eating utensils, after touching animals, and after diapering, nursing, or feeding infants.

GROWTH PHASES OF MICROORGANISMS

Microbial cells have a growth cycle with five phases: lag phase, logarithmic growth phases, stationary growth phase, accelerated death phase, and reduced death phase. Figure 2.1 shows a typical microbial growth curve.

Lag Phase

The lag phase is an adjustment or adaptation period. After the food has been contaminated, microbes take a while to get used to the environment. At first, the stress of the new location may cause a small decrease in the number of microbes. But then they begin to grow slowly. Low temperature extends the lag phase. Cold temperatures reduce growth of microorganisms so that each generation takes longer to multiply. Decreasing the number of microbes that contaminate food, equipment, or buildings also slows their growth. Good sanitation and hygiene lower the initial

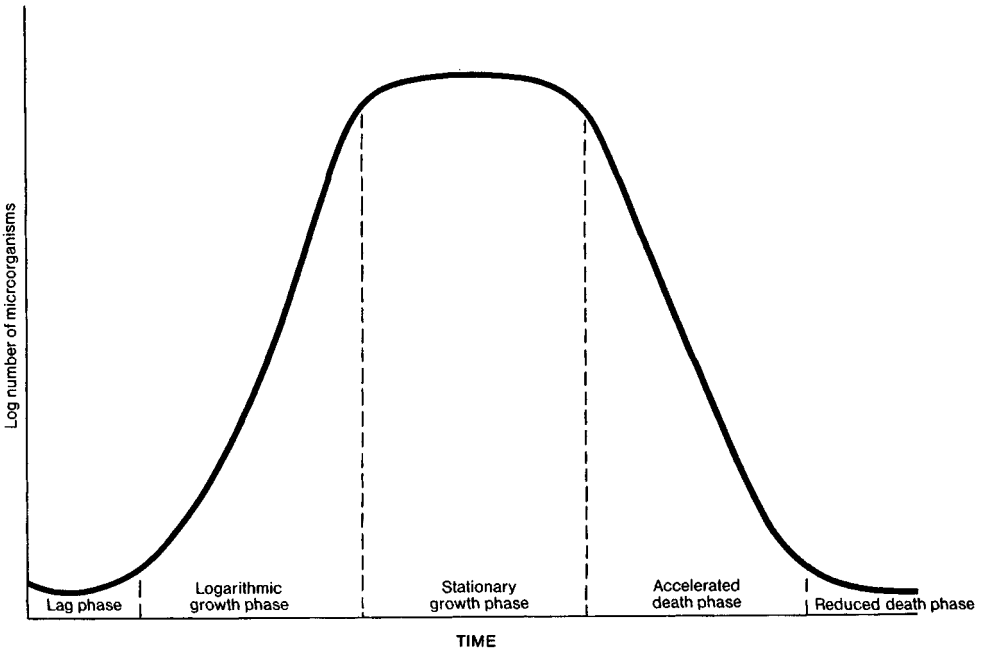


FIGURE 2.1. Typical growth curve for bacteria.

number of microorganisms and can also help to extend the lag phase. Figure 2.2 shows how temperature and contamination levels can affect growth of microbes.

Logarithmic Growth Phase

Bacteria multiply by binary fission, which means that all the cell parts duplicate inside the cell, and then the cell splits into two identical daughter cells. The logarithmic growth phase may last for 2 hours to several hours. The number of microorganisms and environmental factors, such as nutrients and temperature, affect the logarithmic growth rate. Because the population doubles with each generation, and a generation may be short as 20 minutes, it is possible for one cell to become 262,144 cells in just 6 hours!

Stationary Growth Phase

Growth slows down when the environment becomes less ideal because of competition for nutrients, temperature changes, buildup of waste products, or other microbes growing in the same food. This phase usually lasts for between 24 hours and 30 days, depending on the nutrient supply and other factors in the environment.

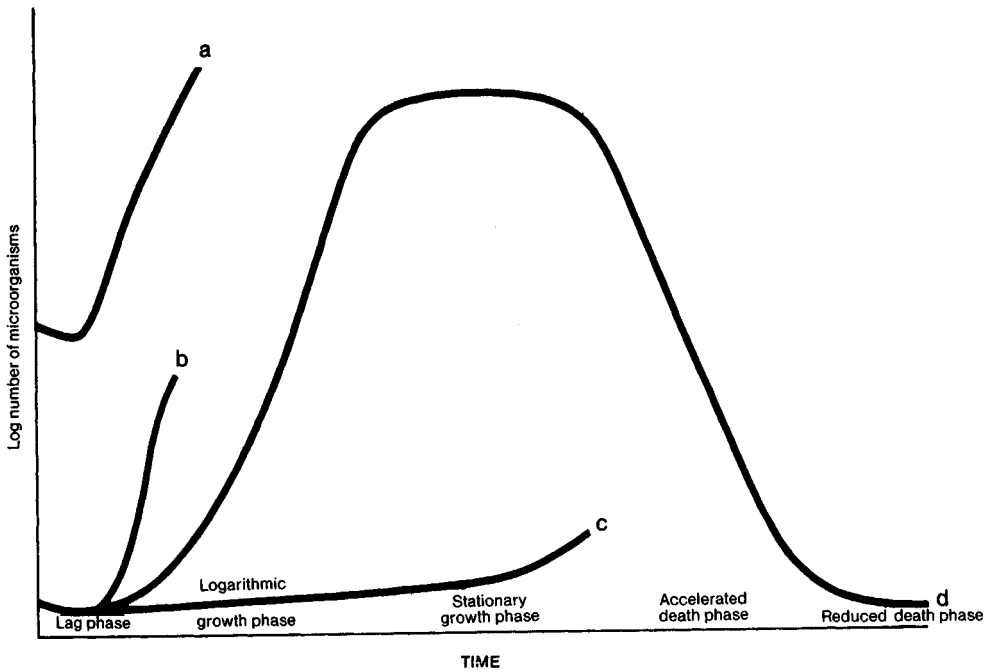


FIGURE 2.2. The effect of amount of initial contamination and length of lag phase on the growth curve of microorganisms: (a) high initial contamination and poor temperature control (short lag phase), (b) low initial contamination and poor temperature control (short lag phase), (c) low initial contamination and rigid temperature control (long lag phase), and (d) typical growth curve.

Accelerated Death Phase

Lack of nutrients, buildup of waste products, and competition from other microbe populations eventually cause rapid death of microbial cells. This phase also lasts between 24 hours and 30 days, depending on the temperature, nutrient supply, type of microbe, age of the microorganisms, use of sanitizers, and other microorganisms in the food.

Reduced Death Phase

When the accelerated death phase reduces the population number, the death rate slows down. After this phase, the organism is decomposed or sterilized, or another microorganism breaks it down.

WHAT CAUSES MICROBES TO GROW?

Temperature

Different microbes each have optimum, minimum, and maximum growth temperatures. Each type of microorganism grows fastest at its optimum temperature, but still

grows at any temperature between its minimum and maximum. The further the temperature is from the optimum temperature for that microorganism, the more slowly it grows. The environmental temperature controls the growth rate and which microorganisms will grow best. For example, a change in temperature of only a few degrees may encourage different organisms to grow in a food. Therefore, small temperature changes can cause different types of food spoilage and foodborne illness. Controlling the temperature is an important way to reduce microbial growth.

Most microorganisms grow best between 14°C (58°F) and 40°C (104°F), although some will grow below 0°C (32°F), and others will grow at temperatures up to 100°C (212°F). Sometimes microbes are classified by their optimal growth temperature:

1. *Thermophiles* prefer hot temperatures above 45°C (113°F), e.g., *Bacillus stearothermophilus, Bacillus coagulans,* and *Lactobacillus thermophilus.*
2. *Mesophiles* prefer medium temperatures between 20°C and 45°C (68°F and 113°F), e.g., most lactobacilli and staphylococci.
3. *Psychrotrophs* tolerate cold temperatures below 20°C (68°F), e.g., *Pseudomonas* and *Moraxella-Acinetobacter.*

Each of these three categories includes some bacteria, molds, and yeasts. Molds and yeasts tend to prefer cooler temperatures than bacteria. When the temperature is below about 5°C (40°F) (refrigerator temperature), microorganisms grow slowly, and many pathogens (disease-causing microbes) stop growing.

Oxygen

Some microorganisms must have oxygen to grow, some grow only without oxygen, and some grow either with or without oxygen. Microorganisms that require oxygen are called *aerobic* microorganisms (e.g., *Pseudomonas* spp.). Microbes that grow without oxygen are called *anaerobic* microorganisms (e.g., *Clostridium* spp.). Those that grow with or without oxygen are called *facultative* microorganisms (e.g., *Lactobacillus* spp.).

Relative Humidity

All microorganisms need water to grow. If the air is humid, water vapor condenses on food, equipment, walls, and ceilings. Moist surfaces encourage microbes to grow. If the air is dry (low relative humidity), microbes are less likely to grow. Bacteria need a higher relative humidity than yeasts and molds. The best relative humidity level for bacteria is 92% or more. Yeasts prefer 90% relative humidity or more. Molds thrive best when the relative humidity is between 85% and 90%.

Water Activity (A_w)

Microbes need water. If less water is available, microbial growth is slower. However, the total amount of water does not affect microbial growth—it is the amount of water that is actually available for the microbes to use that is important. Water activity (A_w)

is the amount of water that is available to microorganisms. Many microorganisms prefer an A_w of .99, and most need an A_w higher than .91 to grow. Relative humidity (RH) and A_w are related (RH = $A_w \times 100$). Therefore, an A_w of .95 is about the same as an RH of 95%. Water activity refers to the availability of water in a food or beverage. Relative humidity refers to the availability of water in the atmosphere around the food or beverage. Most food products have an A_w of about .99. Bacteria need a high A_w, molds need a lower A_w, and most yeasts are between the two. Most spoilage bacteria cannot grow if A_w is below .91, but molds and yeasts can grow when A_w is .80 or lower. Therefore, molds and yeasts are more likely to grow on dried-out foods or surfaces than are bacteria.

In some climates, a high relative humidity may hydrate (dampen) the top layer of foods and allow microorganisms to grow, even if the rest of the food has a low water activity. Good packaging stops the humid air from reaching the food so that microorganisms cannot start to grow.

pH

The pH value shows how acid or alkaline something is; pH ranges from 0.0 to 14.0 (the number is $\log_{10}$ of the reciprocal of the hydrogen ion concentration). pH 7.0 is neutral, pH values below 7.0 are acid, and pH values above 7.0 are alkaline. Most microorganisms prefer a neutral pH. Yeasts can grow in acid and do best when pH is 4.0 to 4.5. Molds can tolerate a wider range of pH (2.0–8.0), although they usually grow best in an acid pH. Molds can grow when conditions are too acidic for bacteria and yeasts. Bacteria grow best when the pH is close to neutral. But some acid-loving (acidophilic) bacteria can grow on food or debris at a pH as low as 5.2. Below pH 5.2, microbial growth is very slow.

Nutrients

Microbes need more than water and oxygen. They also need sources of energy (carbohydrates, fats, or proteins), nitrogen, minerals, and vitamins to grow. Most microbes get nitrogen from amino acids (broken-down proteins) or other chemicals that contain nitrogen (e.g., urea). Molds can use whole proteins, carbohydrates, and fats because they have enzymes that can digest them. Many bacteria can also break down nutrients, but most yeasts need nutrients in simple, broken-down forms. All microorganisms need minerals, but different microbes need different amounts and kinds of vitamins. Molds and some bacteria can make enough B vitamins for themselves, but other microorganisms need a supply of B vitamins.

Inhibitors

Inhibitors slow the growth of microbes. Substances or things that slow the growth of bacteria are called *bacteriostats*. Those that kill bacteria are called *bactericides*. Sometimes food processors add bacteriostatic substances, such as nitrites, to foods to

discourage growth of bacteria. Most bactericides are used to decontaminate foods or to sanitize equipment, utensils, and rooms after they have been cleaned (see Chap. 6).

Biofilms

Biofilms were discovered in the mid-1970s. They are very small colonies of bacteria that stick to a surface using a matrix or glue that traps nutrients and other microorganisms. A biofilm is like a beachhead for bacteria and is not easy to get rid of with sanitizers. When a microorganism lands on a surface, it uses filaments or tendrils to attach itself. The organisms produce a sticky substance that cements the bacteria in place within a few hours. A biofilm builds up several layers of the matrix, populated with microbes such as *Salmonella, Listeria,* and *Pseudomonas.* Eventually the biofilm becomes a tough layer that has to be scraped off. A surface can be cleaned and sanitized, but a well-established biofilm has layers of microorganisms that are protected from the sanitizer. It is easy for chunks of the outer layers of the biofilm to come off when food rubs against it. This contaminates the food.

Biofilms are very hard to remove during cleaning. *Pseudomonas* and *Listeria monocytogenes* are microbes that cause problems because they are protected by biofilms. Heat seems to be more effective than chemical sanitizers for destroying biofilms. Also, it is easier to clear biofilms off Teflon than off stainless-steel surfaces. Water-soluble sanitizers—such as caustics, bleaches, iodophors, phenols, and quaternary ammonium sanitizers—do not get into biofilms. Government regulations and industry guidelines do not cover removal of biofilms. Cleaning employees may need to use a sanitizer at 10 to 100 times the normal strength for it to work. It is very important to rinse the equipment or area carefully after using these high concentrations of sanitizers so that sanitizer residues do not contaminate the food.

Relationship Between Microbial Load, Temperature, and Time

At lower temperatures, the generation interval (time for one bacterial cell to become two cells) is longer. At temperatures below 4°C (40°F), the generation interval is very long. Figure 2.2 shows how temperature affects microbial growth. For example, freshly ground beef usually contains about 1 million bacteria per gram (a gram is about 1/28 of an ounce). When the population reaches about 300 million per gram, the meat becomes slimy and smells "bad." If the meat is more contaminated to begin with or if the temperature is high enough so that bacteria multiply rapidly, the food will spoil more quickly. For example, ground beef that contains 1 million bacteria per gram can be stored for about 28 hours at 15.5°C (60°F) before it is spoiled. At normal refrigerator temperatures (–1–4°C, 30–40°F), it can be stored for more than 96 hours.

HOW FOOD IS SPOILED

Food is spoiled when it is unfit for humans to eat. Spoilage usually means that microorganisms have decomposed or putrefied the food. Food that is unsafe does not

necessarily look or smell spoiled.

Physical Changes

Physical changes are usually more obvious than chemical changes. Spoilage causes physical characteristics such as color, texture, thickness, smell, and flavor to change. Food spoilage is either aerobic or anaerobic, depending on whether oxygen is available. Aerobic spoilage by molds is normally only on the surface of the food, where oxygen is available. Moldy surfaces of foods such as meats and cheeses can be trimmed off, and the rest can be safe to eat. If bacteria grow extensively on the surface, toxins may get inside the food. Anaerobic spoilage happens in sealed containers or in the middle of foods where oxygen is not available. Facultative or anaerobic bacteria spoil food by souring, putrefaction, or tainting. When foods go sour, the microorganisms may release gases that make cans or containers swell.

Chemical Changes

The foods themselves (and the microorganisms that contaminate them) contain enzymes that break down proteins, fats, carbohydrates, and other large molecules into smaller and simpler chemicals. As the number of microbes increases, they start to break down the food. The texture of the food may become soft or slimy and the color may change. Without oxygen, protein may be broken down to a variety of sulfur-containing compounds, which usually smell offensive (like the smell of a bad egg).

FOODBORNE ILLNESS

The United States has the safest food supply in the world. But microorganisms in food still cause about 25 million cases of foodborne illness and 16,000 deaths each year. In the United States, the annual cost of foodborne illness and death averages about $3,000 per person, or about $75 billion. Each death related to foodborne illness costs about $42,300, including insurance and other expenses.

Foodborne Disease

A foodborne disease is any illness caused by eating food. A foodborne disease outbreak is when two or more people have the same illness, which usually involves the stomach and/or intestines, after eating the same food. Bacteria cause about two-thirds of all foodborne illness outbreaks. The incidence of foodborne diseases is not known because so many cases are not reported. But between 6 and 50 million Americans probably become ill each year because of microorganisms in food. *Salmonella* spp. alone causes between 200,000 and 1 million cases each year. About 9,000 people die each year from foodborne illness, and the cause is not known in about 60% of the 200 foodborne disease outbreaks that are reported each year.

Salmonella spp., *Staphylococcus aureus, Clostridium perfringens, Clostridium botulinum, Listeria monocytogenes, Yersinia enterocolitica, Escherichia coli, Vibrio* spp., and *Campylobacter* spp. are all carried in food and could be responsible for illness. Many different home-cooked and commercially prepared foods have caused outbreaks, but foods of animal origin—such as poultry, eggs, red meat, seafood, and dairy products—are most often responsible.

Food Poisoning

It is helpful to understand the names of different types of foodborne illness.

- *Food poisoning* is an illness caused by eating food that contains toxins made by microbes or chemical poisons.
- *Food intoxication* is food poisoning caused by toxins from bacteria.
- *Chemical poisoning* is food poisoning caused by chemicals (such as polychlorinated biphenyls) that are in the food by accident. (Illnesses caused by microorganisms are more common than those caused by chemicals.)
- *Food infections* are illnesses that are not caused by toxins produced by bacteria but are due to eating infectious microorganisms, such as bacteria, viruses, or parasites.
- *Food toxicoinfections* are foodborne illnesses caused from a combination of food intoxication and food infection. Disease-causing bacteria grow in the food, and the patient eats large numbers with the food. The pathogen continues to grow in the intestine and produces a toxin that causes illness symptoms.
- *Psychosomatic food illness* is illness caused by the mind after seeing someone else sick or after seeing a foreign object, such as an insect or rodent, in a food product.

Food producers need up-to-date knowledge of production, harvesting, and storage techniques to evaluate the quality and safety of raw materials. They must understand the design, construction, and operation of food equipment to be able to control the processing, preservation, preparation, and packaging of food products. Food products are vulnerable to contamination and need to be safeguarded to prevent food poisoning.

Staphylococcal Food Poisoning

This is caused by eating the toxin produced by *Staphylococcus aureus*. This facultative, non-spore-forming microorganism produces an enterotoxin that causes irritation of the stomach and intestines. Staphylococcal food poisoning is rarely fatal, but it can cause vomiting and diarrhea. If it does cause death, it is usually because the patient also has other illnesses. The microorganisms that cause staphylococcal food poisoning are very common and can be carried by healthy people. About half of humans are carriers. Food is usually contaminated when infected food handlers touch it. Potato salad, custard-filled pastries, dairy products (including cream), poultry, cooked ham, and tongue are the most common food sources of staphylococcal

food poisoning. If the temperature is ideal and the food is heavily contaminated, staphylococci can multiply enough to cause food poisoning before the color, flavor, or aroma of the food changes. Heating at 66°C (150°F) for 12 minutes destroys *Staphylococcus aureus* organisms, but the toxin they produce is not destroyed by normal cooking or processing.

Salmonella Food Poisoning

Salmonellosis (foodborne illness from *Salmonella*) usually causes nausea, vomiting, and diarrhea, because the toxins irritate the walls of the intestines. About 1 million of these microorganisms have to be eaten by healthy people to cause an infection. It takes fewer organisms to cause salmonellosis in people who are already ill. The time between eating the food and developing symptoms is generally longer than for staphylococcal food poisoning. Salmonellosis rarely causes death, but deaths may occur if the patient is an infant, elderly, or already sick from other illnesses. Patients with acquired immune deficiency syndrome (AIDS) are susceptible to this foodborne illness.

Although salmonellae can be found in animal products, most meat is contaminated by handlers during processing. Heat processes that kill *Staphylococcus aureus* will destroy most species of *Salmonella.* Salmonellosis can usually be blamed on poor sanitation and temperature abuse, because most strains of salmonellae cannot grow at refrigerator temperatures.

Clostridium perfringens Food Poisoning

Clostridium perfringens multiplies rapidly in meat items that have been cooked, allowed to cool slowly, and then held for a long time before serving. Large numbers of active bacteria have to be eaten for this foodborne illness to occur.

Proper sanitation and refrigeration of foods at all times, especially of leftovers, can usually prevent an outbreak of *C. perfringens* foodborne illness. Living microorganisms are destroyed when leftover foods are reheated to 60°C (140°F).

Botulism

Botulism is a true food poisoning caused by eating a toxin produced by *Clostridium botulinum* when it grows in food. The toxin is extremely potent (the second most powerful biological poison for humans) and affects the victim's peripheral nervous system. About 60% of those infected die from respiratory failure. Those who recover may need respiratory treatment for the rest of their lives. Table 2.1 shows the symptoms, incubation time, foods involved, and prevention of botulism and other common food poisonings.

Because *C. botulinum* can be found in soil, it is also found in water. Therefore, seafoods are more common sources of botulism than are animal foods. But the most common sources of botulism are home-canned vegetables and fruits with a low-to-

TABLE 2.1. Characteristics of Foodborne Illnesses

Illness	Causative Agent	Symptoms	Average Time Before Onset of Symptoms	Foods Usually Involved	Preventive Measures
Botulism	Toxins produced by *Clostridium botulinum*	Impaired swallowing, speaking, respiration, and coordination; dizziness and double vision	12–48 hours	Canned low-acid foods, including canned meat and seafood, smoked and processed fish	Proper canning, smoking, and processing procedures; cooking to destroy toxins, proper refrigeration and sanitation
Staphylococcal foodborne illness	Enterotoxin produced by *Staphylococcus aureus*	Nausea, vomiting, abdominal cramps due to gastroenteritis (inflammation of the lining of the stomach and intestines)	3–6 hours	Custard- and cream-filled pastries, potato salad, dairy, ham, tongue, and poultry	Pasteurization of susceptible foods, proper refrigeration and sanitation
Clostridium perfringens foodborne illness	Toxin produced by *Clostridium perfringens*	Nausea, occasional vomiting, diarrhea, and abdominal pain	8–12 hours	Cooked meat, poultry, and fish held at nonrefrigerated temperatures for long periods of time	Prompt refrigeration of unconsumed cooked meat, poultry, or fish; maintain proper refrigeration and sanitation
Salmonellosis (food infection)	Infection produced by ingestion of any of over 1,200 species of *Salmonella* that can grow in the gastrointestinal tract of the consumer	Nausea, vomiting, diarrhea, fever, abdominal pain; may be preceded by chills and headache	6–24 hours	Insufficiently cooked or warmed-over meat, poultry, eggs, and dairy products; these are especially susceptible when kept refrigerated for a long time	Cleanliness and sanitation of handlers and equipment, pasteurization, proper refrigeration and packaging

Table 2.1. *(Continued)*

Illness	Causative Agent	Symptoms	Average Time Before Onset of Symptoms	Foods Usually Involved	Preventive Measures
Ttrichinosis (infection)	*Trichinella spiralis* (nematode worm found in pork)	Nausea, vomiting, diarrhea, profuse sweating, fever, muscle soreness	2–28 days	Insufficiently cooked pork and products containing pork	Thorough cooking of pork to an internal temperature of 59°C (138°F) or 77°C (150°F) or higher with microwave cooking; freezing storage of uncooked pork at minus 15°C (60°F) or lower, for a minimum of 20 days; avoid feeding pigs raw garbage
Aeromonas foodborne illness	*Aeromonas hydrophila*	Gastroenteritis	—	Water, poultry, red meats	Sanitary handling, processing, preparation, and storage of foods; store foods below 2°C (35°F)
Campylobacteriosis foodborne illness	*Campylobacter* spp.	Diarrhea, abdominal pain, cramping, fever prostration, bloody stools, headache, muscle pain, dizziness, and, rarely, death	3–5 hours	Poultry and red meats	Sanitary handling, processing, preparation, and storage of muscle foods
Listeriosis	*Listeria monocytogenes*	Meningitis or meningoencephalitis, listerial septicemia (blood poisoning), fever, intense headache, nausea,	1–7 days	Milk, coleslaw, cheese, ice cream, poultry, red meats	Avoid consumption of raw foods and contact with infected animals; store foods below 2°C (35°F)

Table 2.1. *(Continued)*

Illness	Causative Agent	Symptoms	Average Time Before Onset of Symptoms	Foods Usually Involved	Preventive Measures
		vomiting, lesions after direct contact, collapse, shock, coma; mimics influenza, interrupted pregnancy, stillbirth; 30% fatality rate in infants and immuno-comprised children and adults			
Yersiniosis	*Yersinia enterocolitica*	Abdominal pain, fever, diarrhea, vomiting, skin rashes for 2–3 days, and, rarely, death	24–36 hours	Dairy products, raw meats, seafoods, fresh vegetables	Sanitary handling, processing, preparation, and storage of foods
Escherichia coli 0157:H7 infection	Enterohemmorhagic *Escherichia coli* 0157:H7	Hemorrhagic colitis, hemolytic uremic syndrome with 5–10% acute mortality rate, abdominal pain, vomiting, anemia, thrombocytopenia, acute renal injury with bloody urine, seizures, pancreatitis	—	Ground beef, dairy products, raw beef, water, apple cider, mayonnaise	Sanitary handling, irradiation, cooking to 65°C (149°F) or higher

medium acid content. Recent outbreaks have involved improperly held baked potatoes and grilled onions and some flavored oils. Because *C. botulinum* is anaerobic, it can also contaminate canned and vacuum-packaged foods that are sealed off from oxygen. It is important not to eat foods in swollen cans because the gas produced by the organism causes the swelling. However, some strains of *C. botulinum* do not produce gas. Heating to at least 83°C (180°F) for 30 minutes during processing is enough to protect smoked fish.

Proper sanitation, proper refrigeration, and thorough cooking are essential to prevent botulism. The toxin is destroyed by heat, but the bacterial spores are only destroyed by severe heat treatment. Heating at 85°C (185°F) for 15 minutes destroys the toxin. Table 2.2 shows the combinations of temperatures and times needed to destroy the spores completely.

TABLE 2.2. Temperatures and Times Required to Destroy Completely *Clostridium botulinum* Spores

Temperature, °C	°F	Time, min
100	212	360
105	220	120
110	230	36
115	240	12
120	248	4

Campylobacter

The intestines of wild and domestic animals often contain this microbe. *Campylobacter* is now the greatest cause of foodborne illness in the United States. It is one of the most frequent causes of bacterial diarrhea and other illnesses, and it may also cause ulcers. *Campylobacter* causes veterinary diseases in poultry, cattle, and sheep, and is quite common on raw poultry. *Campylobacter* lives in the intestines of cattle, sheep, swine, chickens, ducks, and turkeys. Animal feces contain the microorganism, so muscle foods (meat and poultry) can be contaminated during slaughtering if sanitary methods are not used. Milk, eggs, and water that have been in contact with animal feces may also contain *C. jejuni*. Red meat is less likely to contain *C. jejuni* than is poultry. It is not likely that *Campylobacter* spp. can be eliminated from domestic animals because it is so widespread.

Campylobacteriosis (foodborne illness from *Campylobacter*) is at least twice as common as salmonellosis. Each year the 4 million cases in the United States cost over $2 billion. The symptoms of foodborne illness from *Campylobacter* vary. People

with a mild case may not have any signs of illness but excrete the microorganism in their feces. People with a severe case may have muscle pain, dizziness, headache, vomiting, cramping, abdominal pain, diarrhea, fever, prostration, and delirium. Patients may find blood in their stool after 1 to 3 days of diarrhea. The illness usually lasts from 2 to 7 days. Death is rare, but it can happen.

Campylobacter outbreaks are most common in children over 10 years old and in young adults, although all age groups are affected. Symptoms and signs of *C. jejuni* infection are similar to those caused by other illnesses of the stomach and intestines. Symptoms may begin between 1 and 11 days after eating contaminated food, but illness usually develops 3 to 5 days after eating this microbe.

Sanitary handling and cooking animal foods properly are the best ways to control *Campylobacter.* Heating contaminated foods to an internal temperature of 60°C (140°F) and holding them at this temperature for several minutes for beef and about 10 minutes for poultry easily destroys *C. jejuni.*

Listeria

The Centers for Disease Control have estimated that 2,000 cases of listeriosis (foodborne illness caused by *Listeria* spp.) occur in the United States each year. A survey during 1992 found that this illness causes about 425 deaths per year.

L. monocytogenes lives in the intestines of over 50 domestic and wild species of animals and birds, including sheep, cattle, chickens, and swine, as well as in soil and decaying vegetation. This microorganism is also found in the air, stream water, sewage, mud, trout, crustaceans (e.g., crabs), houseflies, ticks, and the intestines of human carriers who have symptoms. This pathogen has been a problem in most foods, from chocolate and garlic bread to dairy products, meat, and poultry. Processed milk, soft unpasteurized cheeses, and other dairy products are the most common food sources of *L. monocytogenes,* but vegetables can also be contaminated if they have been fertilized with manure from infected animals. Elimination of *Listeria* is not practical and may be impossible, but food producers and consumers can control its survival.

This microbe grows best at 37°C (98°F), but it can grow at temperatures between 0°C (32°F) and 45°C (113°F). Therefore, it can grow in the refrigerator. It grows twice as fast at 10°C (50°F) as at 4°C (40°F) and will survive freezing, but is usually destroyed at temperatures above 61.5°C (142°F).

Listeriosis is most common in pregnant women, infants, adults over 50 years old, people who are sick from another disease, and other people with reduced resistance or immunity. In adults, the most common symptoms are meningitis or meningoencephalitis. Symptoms may be mild and influenza-like. If a pregnant woman is infected with listeriosis, the fetus may also be infected. This may cause a miscarriage or a stillborn child. If the infant survives, it may be born with septicemia or may develop meningitis. About 30% of newborn infants who get listeriosis die, and about 50% die if they get the infection in the first 4 days after birth.

Listeriosis is very dangerous to persons with acquired immunodeficiency syndrome (AIDS). AIDS severely damages the immune system, so patients are more likely to get a foodborne illness such as listeriosis.

Listeria monocytogenes can stick to food contact surfaces and form a biofilm that is hard to remove during cleaning. These microbes can contaminate food ingredients used in food processing, so the pathogen is constantly brought into the processing plant. A Hazard Analysis Critical Control Point (HACCP) and other quality assurance practices are the best ways to control this pathogen.

This microorganism is usually passed on by eating contaminated food, but it can also be passed by person-to-person contact. People who have been in contact with infected animals, soil, or feces may develop sores on their hands and arms.

About two-thirds of home refrigerators may contain *Listeria* spp. The best way to prevent listeriosis is to avoid ingesting raw (unpasteurized) milk, raw meat, and foods made from contaminated ingredients. Pregnant women should be especially careful to avoid contact with infected animals. Food processors must use a strict sanitation program and HACCP program to control *Listeria.*

Yersinia enterocolitica

Yersinia enterocolitica lives in the intestines and feces of wild and domestic animals. It can also live in raw foods of animal origin and nonchlorinated water from wells, streams, lakes, and rivers. This microorganism also seems to be transmitted from person to person.

Y. enterocolitica can multiply in the refrigerator, but more slowly than at room temperature. This microorganism is destroyed at temperatures over 60°C (140°F). When it is found in processed foods, it is usually occurs because they have been contaminated after cooking. *Y. enterocolitica* has been found in raw or rare red meats and poultry; dairy products such as milk, ice cream, cream, eggnog, and cheese curds; most seafoods; and fresh vegetables.

Not all types of *Y. enterocolitica* cause illness. Yersiniosis (foodborne illness caused by *Yersinia enterocolitica*) is most common in children and teenagers, although adults can get it too. The symptoms (fever, abdominal pain, and diarrhea) normally begin 2 to 7 days after eating the contaminated food. Patients may also vomit and have skin rashes. The abdominal pain caused by yersiniosis is similar to appendicitis. In the past, some children have had their appendix removed because of an incorrect diagnosis.

The illness normally lasts 2 to 3 days, although patients may have mild diarrhea and abdominal pain for 1 to 2 weeks. Death is rare, but can happen if the patient has complications. Proper sanitation in food processing, handling, storage, and preparation is the best way to prevent yersiniosis.

Escherichia coli 0157:H7

Recently, this pathogen has caused serious outbreaks of hemorrhagic colitis and hemolytic uremic syndrome. This pathogen is found in dairy products, water, apple cider, mayonnaise, and raw beef, but the most common cause of outbreaks is undercooked ground beef. Dairy cows can carry *E. coli* 0157:H7. Feces of cattle may contain

this microorganism and can contaminate meat during processing. It is important to monitor and control slaughtering procedures and meat-processing operations to prevent this pathogen from growing. Beef should be cooked to an internal temperature of 70°C (158°F) to destroy this pathogen. A rigid sanitation program is essential to reduce foodborne illness outbreaks from this microorganism.

Other Bacterial Infections

Other bacterial infections cause foodborne illness. The most common of these infections is caused by *Streptococcus faecalis.* Another common cause of infections is *Escherichia coli.* This can cause "traveler's diarrhea," which can be a problem for people from developed countries when they visit developing countries with poor hygiene. Evisceration (removing the intestines) and cold storage of chickens at 3°C may allow *Aeromonas hydrophila* to grow. Water used to chill the birds and the process of removing the intestines may spread bacteria during broiler processing, which may be why *A. hydrophila* is so common on retail cuts of poultry.

Bacillus cereus is another common bacteria that can cause foodborne illness. *B. cereus* foodborne illness causes diarrhea and abdominal pain beginning 8 to 12 hours after the contaminated food is eaten and lasting for about 12 hours. It may also cause vomiting within 1 to 5 hours after eating contaminated food. *B. cereus* produces the toxin in the food before it is eaten. Heat does not easily destroy the toxin. Rice or fried rice from restaurants or warmed-over mashed potatoes that have been cooled slowly before reheating have caused outbreaks. The best way to control this foodborne illness is to use proper sanitation. Because *B. cereus* spores are airborne, food should be covered during holding, when possible. Restaurants should hold cooked starchy foods above 50°C (122°F) or cool them to less than 4°C (40°F) within 2 hours of cooking to prevent the bacteria from growing and producing the toxin.

Mycotoxins

Mycotoxins are produced by many fungi. The diseases caused by mycotoxins are called *mycotoxicoses.* Mycotoxicoses are not common in humans, but one mycotoxin (aflatoxin) seems to be linked to liver cancer in susceptible populations. Large doses of aflatoxins are very toxic, causing liver damage, intestinal and internal bleeding, and death. Mycotoxins get into food when mold grows on the food or when food producers use contaminated ingredients in processed foods.

Molds that can produce mycotoxins often contaminate foods. *Aspergillus, Penicillium, Fusarium, Cladosporium, Alternaria, Trichothecium, Byssochlamys,* and *Sclerotinia* are important in the food industry. Most foods can be infected by these or other fungi at some point during production, processing, distribution, storage, or display. If mold grows, it may produce mycotoxins. However, not all moldy foods contain mycotoxins. Foods that are not moldy are not necessarily free of mycotoxins, because the toxin can still be there after the mold has disappeared.

Aflatoxin is the most hazardous mycotoxin for humans. It is produced by *Aspergillus flavus* and *A. parasiticus.* These molds are found almost everywhere, and

their spores are easily carried by air currents. They are often found in cereal grains, almonds, pecans, walnuts, peanuts, cottonseed, and sorghum. The microorganisms do not grow unless the food is damaged by insects, not dried quickly, or stored somewhere damp.

Aflatoxicosis causes loss of appetite, lethargy, weight loss, neurological problems, convulsions, and sometimes death. It can also cause liver damage, fluid buildup (edema) in the body cavity, and bleeding in the kidneys and intestines.

The best way to eliminate mycotoxins from foods is to prevent mold growth, insect damage, and mechanical damage during production, harvesting, transporting, processing, storage, and marketing. Mycotoxins are not produced when A_w is below .83 (about 8 to 12% moisture in grain). Therefore, grains should be dried quickly and completely and stored somewhere dry. Photoelectric eyes can examine grain and automatically remove discolored kernels that may contain aflatoxins. This equipment is used in the peanut industry to avoid difficult, boring, and expensive hand sorting.

Are There More Pathogens in Food Than There Used to Be?

Pathogens are found in food more often than in the past, but it may be because scientists have developed better methods to detect pathogens, rather than because foods are more contaminated. Other reasons why pathogens are found more often include the following:

1. *Changes in eating habits.* "Organic" products may seem healthy but can be unsafe. An outbreak of listeriosis in Canada was linked to coleslaw that was made from cabbage fertilized with sheep manure.
2. *More awareness of hazards, risks, and hygiene.* Better collection of data and use of computers have helped public health officials recognize foodborne listeriosis.
3. *Changes in the population.* Ill people are kept alive much longer. These people are more likely to be infected. Tourists and immigrants may spread certain diseases worldwide.
4. *Changes in food production.* Raw materials are often produced in large amounts. Large batches can easily create places for microorganisms to grow (e.g., because the center takes longer to cool to refrigerator temperature after cooking). A large batch of food can spread contamination even further when it is used as ingredients in other foods.
5. *Changes in food processing.* Vacuum packages and cold storage allow facultative microorganisms and psychrotrophs to survive.
6. *Changes in food storage.* Food processors and consumers can store foods such as vegetables, salads, soft cheeses, and meats for longer in the refrigerator, but this can allow growth of psychrotrophic pathogens such as *Listeria monocytogenes.*
7. *Changes in microorganisms.* Over time, the genetic makeup of microorganisms can change. New strains may develop that have characteristics from different bacteria. For example, a pathogen may become able to grow in cold temperatures, survive cooking, or grow in more acid or alkaline foods.

DESTROYING MICROBES

Regardless of how microorganisms are destroyed, they die at a relatively constant rate in the accelerated death phase (Fig. 2.1). Some things can change the death rate, such as a lethal agent or a mixed population of sensitive and resistant cells. Heat, chemicals, and radiation can all destroy microbes.

Heat

Heat is the most common method of killing spoilage and pathogenic bacteria in foods. Heat processing is a way to cook food and destroy microorganisms at the same time. Researchers work hard to find the ideal temperature and length of cooking time to destroy harmful microorganisms without overcooking the food. The amount of time it takes to completely sterilize a liquid containing bacterial cells or spores at a given temperature is the *thermal death time* (TDT). The TDT depends on the microorganism, the number of cells, and what it is growing on.

Another way to measure destruction of microbes is *decimal reduction time* (D value). This is the number of minutes it takes to destroy 90% of the cells at a given temperature.

Chemicals

Many chemical compounds that destroy microorganisms should not be used to kill bacteria in or on food. Food processors use chemicals to sanitize equipment and utensils that could contaminate food. These chemicals are rinsed off, so they cannot contaminate food. Sanitizing using heat has become more expensive, so the food industry uses chemical sanitizers more often.

Radiation

Microbiologists do not fully understand how radiation destroys microbes. It seems to inactivate parts inside the cell as they absorb its energy. When radiation inactivates a cell, it cannot divide and grow.

INHIBITING GROWTH OF MICROBES

Food processors use mild versions of methods used to kill microorganisms to inhibit or slow down growth of microbes. Heating at lower temperatures or for shorter times, low-dose irradiation, or treatment with diluted chemicals (such as food preservatives) can injure microorganisms and slow their growth without killing them. Injured microorganisms have a longer lag phase; do not tolerate extreme heat, cold, acid, or alkali so well; and are more sensitive to less-than-ideal conditions. Microbes are more easily inhibited by two inhibitors at the same time, such as irradiation plus heat, or heat plus chemicals.

Freezing

Freezing and thawing kills some microbes. Those that survive freezing cannot grow during frozen storage. But freezing is not a practical way to reduce the number of microbes. Also, once the food is thawed, the microorganisms that survive freezing will grow as quickly as those that have not been frozen. Refrigeration works well with other methods of inhibition, such as preservatives, heat (pasteurization), and irradiation.

Chemicals

Chemicals that reduce A_w to a level that prevents growth of most bacteria can control bacteria. Salt and sugar are examples of chemicals that reduce A_w. When they are used to preserve foods, they are used in much greater amounts than those used to flavor and season foods. Chemical preservatives also prevent growth of microbes in foods by reducing the pH, altering the A_w, or interfering with the microbe's metabolism.

Dehydration

Dehydration also reduces the A_w to a level that prevents microbial growth. Dehydration is most effective when foods are also treated with another inhibitor, such as salt or refrigeration.

Fermentation

Fermentation produces desirable flavors and can control microbial growth. Fermentation produces acid that lowers the pH of the food. A pH below 5.0 restricts growth of spoilage microorganisms. Foods that are fermented and heated can be canned or bottled to prevent spoilage by aerobic yeasts and molds (e.g., pickles).

TESTS TO DETECT AND COUNT MICROBES

Various methods can detect growth of microbes in foods. Food scientists use different methods depending on what they need to know, the type of food, and the type of microbe. The American Public Health Association (APHA) and the Association of Official Analytical Chemists (AOAC) have developed standard testing procedures. It is very important to collect representative samples (see Chap. 10). Microbial tests are less accurate and precise than chemical tests. Therefore, food scientists need technical knowledge about microbiology and food products to choose the right test and interpret the results.

Although microbial analysis may not give exact results, it can show whether the sanitation program is working to keep food products safe and equipment, utensils, floors, walls, and other areas clean. Microbial tests can also predict shelf life or how long a food can be kept. Here is a brief summary of the most common microbial tests.

Total Plate Count

This counts the total population of aerobic microorganisms on equipment or food products. The total plate count method assesses contamination from the air, water, equipment surfaces, facilities, and food products. The microbiologist swabs the equipment, walls, or food products. He or she washes the material off the swab into a culture medium (such as standard-methods agar or plate count agar) that supports the growth of all microorganisms. The microbiologist dilutes the sample several times, depending on how many microbes are expected, and places it on a growth medium in a sterile covered plate (petri dish). The incubation temperature may be the same as the storage temperature of the food or the same temperature as the location of equipment or utensils (e.g., 4°C [40°F]) for refrigerator shelves, room temperature for a knife). The result is the number of colonies that grow. This does not show the type of microorganisms, although some colonies look different. Sometimes the bacteria are incubated on special growth media that allow only certain microorganisms to grow and be counted.

Press Plate Technique

This is also called the *contact plate technique.* It is similar to the plate count technique without the swabbing. The microbiologist opens a covered RODAC plate (similar to a petri dish) and presses the growth medium (agar) against the area to be sampled. The incubation process is the same as for the total plate count method. This method is easy and has less chance of error and contamination. The biggest problem is that it can only be used for flat surfaces that are lightly contaminated because the sample is not diluted. Sanitarians use press plates to check whether the sanitation program is effective. The number of colonies that grow show the amount of contamination.

Indicator and Dye Reduction Tests

Many microorganisms produce enzymes during their normal growth. These enzymes can react with indicators, such as dyes. The enzymes react with the dye and cause a color change. The speed of the reaction shows the number of microorganisms in the sample. Sometimes a dye-soaked filter paper is placed right onto a food sample or piece of equipment. The time it takes for the filter paper to change color shows the number of microbes. The biggest problem with this technique is that it does not show the exact amount of contamination. But it is quicker and easier than plate counts and can be a good tool to check the effectiveness of a sanitation program. There are many different test kits for specific organisms.

Direct Microscopic Count

The microbiologist dries and fixes a measured sample of food on a microscope slide. He or she stains it and counts the number of bacteria. Most stains do not distinguish between bacteria that are alive or dead, so this method shows the total number of

microorganisms in a sample. This method gives some information about the type of bacteria in the sample, and the slides can be kept for future reference, but it is not used much because it is easy for analysts to make counting errors, and the method can examine only a small quantity.

Most Probable Number (MPN)

This common method estimates bacterial populations by lining up several tubes of liquid growth medium and adding a sample that has been diluted by different amounts to each tube. If bacteria grow, the medium looks cloudy. The number of microorganisms is shown by how much the sample had to be diluted to prevent growth. This method measures only live bacteria, and the microbiologist can do more tests on the bacteria that grow to identify them.

Impedance Measurements

Impedance can measure microbial metabolism, which is an indirect way to count the number of microbes in a sample. Impedance is the total resistance to the flow of an alternating electric current as it passes through something. Microbiologists measure microbial growth by looking at the changes in impedance measurements over 5 hours. In the future, sanitarians may use impedance as a rapid method of counting microbial load, although it can be used only when the food contains more than 100,000 microbes per gram. At the moment, the equipment costs about $70,000, and each sample costs about $2. The equipment can handle 128 to 512 samples at one time.

Direct Epifluorescence Filter Technique (DEFT)

This is a rapid, direct method of counting microorganisms in a sample. Cells are stained with a fluorescent dye that stains live and inactive bacteria different colors. English researchers developed this method to monitor milk samples, but food scientists also use it for other foods. Sanitarians use DEFT to check dairy foods and meat, beverages, water, and wastewater. Each test takes about 25 minutes, and each sample costs about $1, although costs vary and are based on the quantity purchased.

Salmonella 1-2 Test

This is a rapid screening test for *Salmonella.* If a dark band (immobilization band) forms in the medium, motile *Salmonella* have reacted with flagellar antibodies, and the test is positive.

This test uses a clear plastic device with two chambers. The chambers contain two different media. One contains flagellar antibodies to *Salmonella,* and the sample is added to the other. After about 4 hours of incubation, motile *Salmonellae* move from one chamber to the other and come in contact with the flagellar antibodies. The immobilization band forms after 8 to 14 hours.

CAMP Test

This is a test for *L. monocytogenes*. The sanitarian streaks the sample next to or across a streak of a known bacterium on a blood agar plate. Where the two streaks meet, the metabolic by-products of the two bacteria blend and react to break down the blood (hemolysis) in the plate medium.

Fraser Enrichment Broth/Modified Oxford Agar

Fraser broth encourages growth of *Listeria* and prevents growth of other microorganisms. A specially prepared U-shaped tube contains Fraser broth on both sides and modified Oxford agar in the middle. The sample is placed into one side of the tube. *Listeria* are the only microorganisms that grow and migrate through the modified Oxford agar into the other side of the tube. The pure culture causes black deposits in the Oxford agar and cloudiness in the second branch of Fraser broth.

SUMMARY

- Microbiology is the study of microscopic forms of life.
- Sanitarians need to understand how microorganisms cause food spoilage and foodborne illness so that they can handle foods in a sanitary way.
- Microorganisms cause food spoilage by changing how food looks, tastes, and smells. Foodborne illness happens when people eat food containing pathogenic microorganisms or their toxins.
- A sanitation program controls the number of microorganisms on equipment, processing plants, and food.
- Microorganisms have a growth pattern that looks like a bell curve. They grow and die at a logarithmic rate. The factors that affect growth of microbes include temperature, oxygen availability, water availability, pH, nutrients, and inhibitors.
- Microorganisms produce enzymes that break down proteins, fats, carbohydrates, and other molecules in food into simpler compounds. This breakdown causes food spoilage. Microorganisms that cause foodborne illness include *Staphylococcus aureus, Salmonella* spp., *Clostridium perfringens, Clostridium botulinum, Campylobacter* spp., *Listeria monocytogenes, Yersinia enterocolitica,* and molds that produce mycotoxins.
- The best ways to destroy microbes are heat, chemical sanitizers, and irradiation. The best ways to inhibit growth of microbes are refrigeration, dehydration, and fermentation.
- Sanitarians use several tests to measure the number and type of microbes in food samples and on equipment or other surfaces. These tests show whether the sanitation program is working.

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STUDY QUESTIONS

1. Name the most common types of microorganisms in food, and give one example of each.
2. What do yeasts and molds look like on food?
3. Describe the five phases of microbial growth.
4. Which microorganisms grow best when water activity is low?
5. Name five factors that affect the growth of microorganisms.
6. Why are biofilms a problem in the food industry?
7. What is the difference between physical and chemical food spoilage?
8. Name four bacteria that commonly cause foodborne illness.
9. Which foods are likely to be contaminated with aflatoxin? How should foods be stored so that molds do not produce aflatoxin?
10. What is the difference between destroying and inhibiting microbes?

TO FIND OUT MORE ABOUT MICROORGANISMS IN FOOD

1. Call USDA's Meat & Poultry Hotline, (202) 472-4485, and ask for information about harmful microorganisms that grow in food.

2. Conduct an informal survey among your friends and family. Has anyone had a foodborne illness? Do they know what food caused the illness? What were the symptoms? How long did the illness last? Do they know what microorganism caused the illness? Was the illness reported to CDC?
3. Call a local fast-food chain, and ask what it does to protect its customers from foodborne illness such as *Escherichia coli* in undercooked ground beef.
4. Check your kitchen at home. Are all foods stored at the right temperature? Are foods covered to protect them from insects, moisture, and microorganisms in the air? Use the kitchen quiz "Can your kitchen pass the food safety test?" by Paula Kurtzweil in FDA Consumer, October 1995.